

For full credit explain your reasoning, showing all relevant work.

Exercise 1. Let R be a commutative ring with unity $1 \in R$. Show that if the only ideals of R are R and $\{0\}$, then R is a field.

Exercise 2. Show that the set $J = \{(3m, n) \mid m, n \in \mathbb{Z}\}$ is a maximal ideal of the ring $\mathbb{Z} \times \mathbb{Z}$.

Exercise 3. a) Determine all ring homomorphisms $f: \mathbb{Q} \rightarrow \mathbb{Q}$. Justify your answer.

Note. We do not assume here that a ring homomorphism needs to preserve the unity in a ring.

b) Give an example of a function $f: \mathbb{Q} \rightarrow \mathbb{Q}$ which is a group homomorphism (with respect to addition in $\mathbb{Q}$) but it is not a ring homomorphism.

PRACTICE PROBLEMS

Exercises below are for practice only - do not turn them in for grading.

Practice Exercise 1. a) Let R be an integral domain consisting of finitely many elements. Show that R is a field.

b) Let R be a commutative ring with unity $1 \in R$, which consists of finitely many elements. Let $I \triangleleft R$ be a prime ideal of R . Show that I is a maximal ideal.

Practice Exercise 2. a) Show that the set $J = \{(a, 0) \mid a \in \mathbb{Z}\}$ is a prime ideal of the ring $\mathbb{Z} \times \mathbb{Z}$.

b) Show that the ideal J from part a) is not a maximal ideal in $\mathbb{Z} \times \mathbb{Z}$.

Practice Exercise 3. Determine all ring homomorphisms $f: \mathbb{Z}_{25} \rightarrow \mathbb{Z}_{20}$.

Note. We do not assume here that a ring homomorphism needs to preserve the unity in a ring.

Practice Exercise 4. a) Let $f: R \rightarrow S$ be a ring homomorphism. Show that $\text{Ker}(f) = \{0\}$ if and only if f is 1-1.

b) Let F be a field and let R be ring. Show that if $f: F \rightarrow R$ is a ring homomorphism then either $f(a) = 0$ for all $a \in F$ or f is 1-1.